

Diabetes Prediction Using Support Vector Machine (SVM)

This project focuses on predicting whether a patient is diabetic or non-diabetic using Machine Learning. The model is built using the Support Vector Machine (SVM) algorithm from Scikit-learn. The dataset used is the PIMA Indians Diabetes Dataset.

Project Objectives

Understand supervised learning concepts
Preprocess and standardize medical data
Train an SVM classification model
Evaluate model accuracy
Predict diabetes status for new patient data

Dataset Features

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Glucose concentration
BloodPressure	Blood pressure level
SkinThickness	Skin fold thickness
Insulin	Insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes hereditary function
Age	Patient age
Outcome	0 = Non-Diabetic, 1 = Diabetic

Project Workflow

Load dataset using Pandas
Separate features (X) and target variable (y)
Scale features using StandardScaler
Split data into training and testing sets
Train SVM model with linear kernel
Evaluate model accuracy
Predict diabetes status for new data

Why StandardScaler is Used

Different features in the dataset have different ranges. For example, glucose values can be much larger than BMI values. StandardScaler standardizes the data so all features contribute equally during training. It transforms the data so that the mean becomes 0 and standard deviation becomes 1.

Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification problems. It works by finding the best boundary (hyperplane) that separates diabetic and non-diabetic patients. In this project, a linear kernel is used for classification.

Model Evaluation

The model performance is measured using accuracy score. Training accuracy checks how well the model learned from training data. Testing accuracy checks how well the model performs on unseen data.

Sample Prediction

Sample patient data is converted into a NumPy array, reshaped, standardized, and then passed to the trained model. The model predicts: 0 → Non-Diabetic 1 → Diabetic

Technologies Used

Python Pandas NumPy Scikit-learn Jupyter Notebook / Google Colab

Conclusion

This project successfully demonstrates how Machine Learning can be used in healthcare prediction systems. Using SVM and proper feature scaling, the model can classify diabetic and non-diabetic patients with good accuracy. The project also helps beginners understand the complete ML workflow from preprocessing to prediction.

Project Highlights

End-to-end Machine Learning project Data preprocessing and scaling SVM classification model Accuracy evaluation Real-time prediction workflow